

ANDREW K. SAYDJARI

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PUBLICATIONS

I am an author on **50+ papers** that have **2274+** citations (h-index=19). This includes:

12+ papers as (co-)lead author with 440+ citations

9+ papers with **significant contributions** with 301+ citations

3+ papers led by **students I (co-)advised** (denoted by *)

Most of my papers can be found online on [ADS](#), though citations outside astronomy are missing.
 My ORCID is [0000-0002-6561-9002](https://orcid.org/0000-0002-6561-9002).

Publications as (Co-)Lead Author:

1. Saydjari, A. K. & Green, G. M.
 2026, ApJ, 999, 28. Postdoc
[Correlations between Dust Extinction Features across All Wavelength Scales: From Diffuse Interstellar Bands to R\(V\)](#)
2. Zucker, C. & Saydjari, A. K. & Speagle, J. S.; Schlafly, E. F.; Green, G. M.; Benjamin, R.; Peek, J. E. G.; Edenhofer, G.; Goodman, A.; Kuhn, M. A.; Finkbeiner, D. P.
 2025, ApJ, 992, 39. Postdoc
[A Deep, High-Angular Resolution 3D Dust Map of the Southern Galactic Plane](#)
3. Saydjari, A. K.; Finkbeiner, D. P.; Wheeler, A. J.; Holtzman, J. A.; Wilson, J. C.; Casey, A. R.; Sánchez-Maes, S.; Brownstein, J. R.; Hogg, D. W.; Blanton, M.
 2025, AJ, 169, 167. Postdoc
[Improving Radial Velocities by Marginalizing over Stars and Sky: Achieving 30 m/s RV Precision for APOGEE in the Plate Era](#)
4. Saydjari, A. K.; Uzsoy, A.S.; Zucker, C.; Peek, J. E. G.; Finkbeiner, D. P.
 2023, ApJ, 954, 141. Grad
[Measuring the 8621 Å Diffuse Interstellar Band in Gaia DR3 RVS Spectra: Obtaining a Clean Catalog by Marginalizing over Stellar Types](#)
5. Saydjari, A. K.; Schlafly, E. F.; Lang, D.; Meisner, A. M.; Green, G. M.; Zucker, C.; Zelko, I.; Speagle, J. S.; Daylan, T.; Lee, A.; Valdes, F.; Schlegel, D.; Finkbeiner, D. P.
 2023, ApJS, 264, 28. Grad
[The Dark Energy Camera Plane Survey 2 \(DECaPS2\): More Sky, Less Bias, and Better Uncertainties](#)
6. Saydjari, A. K.; Finkbeiner, D. P.
 2022, ApJ, 933, 155. Grad
[Photometry on Structured Backgrounds: Local Pixel-wise Infilling by Regression](#)
7. Saydjari, A. K.; Finkbeiner, D. P.
 2023, TPAMI, 45, 1716. Grad
[Equivariant Wavelets: Fast Rotation and Translation Invariant Wavelet Scattering Transforms](#)
8. Saydjari, A. K.; Portillo, S. K. N.; Slepian, Z.; Kahraman, S.; Burkhart, B.; Finkbeiner, D. P.
 2021, ApJ, 910, 122. Grad

[Classification of Magnetohydrodynamic Simulations using Wavelet Scattering Transforms](#)

9. **Saydjari, A. K. & Weis, P.;** Wu, S.
2017, Adv. Energy Mat., 7, 1601622. Undergrad
[Spanning the Solar Spectrum: Azopolymer Solar Thermal Fuels for Simultaneous UV and Visible Light Storage](#)
Cover Article: [AEM](#)
10. **Saydjari, A. K.;** Pietron, J. J.; Simpkins, B. S.
2015, Electroanalysis, 27, 1960-1967. HS
[Electrochemical Deposition and Spectroelectro-chemical Response of Bromophenol Blue Films on Gold](#)
11. **Saydjari, A. K.;** Long, J.; Dressick, W.; Simpkins, B. S.
2014, Chem. Phys. Lett., 608, 328-333. HS
[Optical Interference Effect Corrections for Absorbance Spectra of Layer-by-Layer Thin Films Bearing Covalently Bound Dye](#)

Publications with Significant Contributions:

12. *Wiemann, M. L., Melchior, P., **Saydjari, A. K.**
2026, arXiv: 2607.14466 Postdoc
[Interleaved Noise Injection Improves Clean, Corrupted, and OOD Performance](#)
Feedback on experimental design, methods iteration, and manuscript text/figures.
13. *O'Neill, T.J.; **Saydjari, A. K.;** et al. [5 additional co-authors]
2026, arXiv: 2605.24342 Postdoc
[The 3D Structure and Kinematics of the Local Disk-Halo Interface: Intermediate-velocity Clouds are the Minority of High-altitude Clouds in the Solar Neighborhood](#)
Project ideation, planning, and co-advising, along with preparation for kinematic modeling.
14. Li, B.; McKinnon, K. A.; **Saydjari, A. K.;** et al. [6 additional co-authors]
2026, PASP, 138, 044503. Postdoc
[Optimal and Unbiased Fluxes from Up-the-Ramp Detectors under Variable Illumination](#)
Significant advising on statistical tests, data reductions, data products, and manuscript text.
15. *Uzsoy, A. M.; **Saydjari, A. K.;** et al. [38 additional co-authors]
2026, ApJ, 999, 241. Postdoc
[Bayesian Component Separation for DESI LAE Automated Spectroscopic Redshifts and Photometric Targeting](#)
Co-advised the application of MADGICS to DESI spectra (prior refinement, error calibration, etc.)
16. Lesser, O.; **Saydjari, A. K.;** Wesson M.; Yacoby, A.; Oreg, Y.
2021, PNAS, 118, 27. Grad
[Phase-induced topological superconductivity in a planar heterostructure](#)
Designed and ran experiments in Kwant to show applicability for realistic device densities.
17. Alegria, L.D.; Böttcher, C.G.; **Saydjari, A. K.;** Pierce, A.T.; Lee, S.H.; Harvey, S.P.; Vool, U.; Yacoby, A.;
2021, Nature Nanotechnology, 16, 404-408. Grad
[High-energy quasiparticle injection into mesoscopic superconductors](#)
Helped fabricate devices, collected data on dil. fridge I operated, and analyzed data/models.
18. Charboneau, D.J.; Brudvig, G.W.; Hazari, N.; Lant, H.M.C; **Saydjari, A. K.**
2019, ACS Catal., 9, 3228-3241. Undergrad

[Development of an Improved System for the Carboxylation of Aryl Halides through Mechanistic Studies](#)

M.Sc. thesis work. Designed experiments, worked up reactions and NMR, analyzed data.

19. McKee, A.; Solano, M.; **Saydjari, A. K.**; Bennett, C.J.; Hud, N.V.; Orlando, T.M.

2018, ChemBioChem, 19, 1913-1917.

Undergrad

[A Possible Path to Prebiotic Peptides Involving Silica and Hydroxy Acid-Mediated Amide Bond Formation](#)

Cover Article: [ChemBioChem](#)

Designed/ran cycling experiments, analyzed MS/MS spectra, wrote first manuscript.

20. Wang, D.; Wagner, M.; **Saydjari, A. K.**; Wu, S.; Butt, H.

2017, Chem. Eur. J., 23, 2628-2634.

Undergrad

[A Photoresponsive Orthogonal Supramolecular Complex Based on Host–Guest Interactions](#)

Lead quantum chemical simulations determining molecular geometry key to HG interaction.

Publications as a Contributing Author:

21. SDSS Collaboration. [275 additional co-authors including **Saydjari, A. K.**]

2026, arXiv:2607.26149 (submitted to AAS Journals)

Postdoc

[The Twentieth Data Release of the Sloan Digital Sky Survey: First All-Sky BOSS Spectra, eROSITA-SDSS-V Mapper Coordinated Observations, and a Preview of the Local Volume Mapper](#)

22. Thibodeaux, P. [14 additional co-authors including **Saydjari, A. K.**]

2026, arXiv: 2607.22499.

Postdoc

[Payne4GAIN: NLTE Corrections for Red Giants in Milky Way Mapper using H-Band Neural Network Emulators](#)

23. McCallum, L. [11 additional co-authors including **Saydjari, A. K.**]

2026, MNRAS, 548, stag640.

Postdoc

[The radial component of the local Galactic magnetic field in 3D](#)

24. DESI Collaboration [277 additional co-authors including **Saydjari, A. K.**]

2026, AJ, 171, 285.

Postdoc

[Data Release 1 of the Dark Energy Spectroscopic Instrument](#)

25. Hilder, T. [16 additional co-authors including **Saydjari, A. K.**]

2026, ApJ, 997, 339.

Postdoc

[Unified Spectrospatial Forward Models: Spatially Continuous Maps of Weak Emission Lines in the Rosette Nebula with SDSS-V LVM](#)

26. McQuaid, T. [9 additional co-authors including **Saydjari, A. K.**]

2026, ApJ, 1001, 237.

Postdoc

[An SDSS-V 3D Tomographic Na I Map of the Interstellar Medium: An Initial Study Towards the Smith Cloud](#)

27. Zari, E. [19 additional co-authors including **Saydjari, A. K.**]

2025, A&A, 703, A303.

Postdoc

[The large-scale kinematics of young stars in the Milky Way disc: first results from SDSS-V](#)

28. Chandra, V. [30 additional co-authors including **Saydjari, A. K.**]

2026, ApJ, 1000, 283.

Postdoc

[Mapping the Distant and Metal-Poor Milky Way with SDSS-V](#)

29. Aftab, N. [8 additional co-authors including **Saydjari, A. K.**]

2026, AJ, 171, 17.

Postdoc

[The MDW H \$\alpha\$ Sky Survey: Data Release 1](#)

30. Medan, I. [9 additional co-authors including **Saydjari, A. K.**]
2025, AJ, 170, 302. Postdoc
[The Importance of Standardizing Spectra in the Era of Large Spectroscopic Surveys: A Case Study of M Dwarfs in SDSS-V](#)
31. Casey, A. R. [5 additional co-authors including **Saydjari, A. K.**]
2026, ApJ, 998, 192. Postdoc
[A constrained linear model for continuum normalisation of stellar spectra](#)
32. Otto, J. M. [17 additional co-authors including **Saydjari, A. K.**]
2026, AJ, 171, 91. Postdoc
[The Open Cluster Chemical Abundances and Mapping Survey: VIII. Galactic Chemical Gradient and Azimuthal Analysis from SDSS/MWM DR19](#)
33. SDSS Collaboration. [211 additional co-authors including **Saydjari, A. K.**]
2026, ApJS, 285, 9. Postdoc
[The Nineteenth Data Release of the Sloan Digital Sky Survey](#)
34. Kollmeier, J.A. [438 additional co-authors including **Saydjari, A. K.**]
2026, AJ, 171, 52. Postdoc
[Sloan Digital Sky Survey-V: Pioneering Panoptic Spectroscopy](#)
35. Mészáros, S. [27 additional co-authors including **Saydjari, A. K.**]
2025, AJ, 170, 96. Postdoc
[SDSS-V Milky Way Mapper \(MWM\): ASPCAP Stellar Parameters and Abundances in SDSS-V Data Release 19](#)
36. Stone-Martinez, A. [6 additional co-authors including **Saydjari, A. K.**]
2025, AJ, 170, 66. Postdoc
[StarFlow: Leveraging Normalizing Flows for Stellar Age Estimation in SDSS-V DR19](#)
37. Speagle, J. S. & Zucker, C. [17 additional co-authors including **Saydjari, A. K.**]
2025, arXiv:2503.02227 (submitted to AAS Journals) Grad
[Deriving Stellar Properties, Distances, and Reddenings from Photometry and Astrometry with BRUTUS](#)
Provided feedback during method development, and on publication/code-base.
38. Zhou, R. [56 additional co-authors including **Saydjari, A. K.**]
2025, OJAp, 8, 83. Postdoc
[Stellar reddening map from DESI imaging and spectroscopy](#)
Aided in map validation, data-release, reddening methods, and provided internal DESI review.
39. Speagle, J. S. & Zucker, C. [17 additional co-authors including **Saydjari, A. K.**]
2024, ApJ, 970, 121. Grad
[Mapping the Milky Way in 5-D with 170 Million Stars](#)
Provided feedback during method development, and on publication/code-base.
40. Edenhofer, G. [7 additional co-authors including **Saydjari, A. K.**]
2024, A&A, 685, A82. Grad
[A Parsec-Scale Galactic 3D Dust Map out to 1.25 kpc from the Sun](#)
Aided in map validation, data-release, securing compute, and consulted on methods development.
41. Almeida, A. [154 additional co-authors including **Saydjari, A. K.**]
2023, ApJS, 267, 44. Grad
[The Eighteenth Data Release of the Sloan Digital Sky Surveys: Targeting and First Spectra from SDSS-V](#)
Key APOGEE pipeline team member and led dust group DIB catalog and reddening efforts.

42. Cantat-Gaudin, T. et al. [13 additional co-authors including **Saydjari, A. K.**] 2023, A&A, 669, A55. Grad
[An empirical model of the Gaia DR3 selection function](#)
 Created DECaPS2 (and high-resolution coverage maps) that were the underlying comparison used in deriving the empirical selection function.

Non-Refereed

White Papers as (Co-)Lead Author:

43. **Saydjari, A. K.** et al. [24 additional co-authors] 2025, PRIMA GO Science Book Volume 2, 481. Postdoc
[PRIM\(All\): A PRIMA All-Sky, Polarized, Dust Map and Point Source Catalog](#)

White Papers with Significant Contributions:

44. Decleir, M. et al. [18 additional co-authors including **Saydjari, A. K.**] 2026, arXiv:2605.18992. Postdoc
[Summary of Discussion Sessions from "The Dusty Universe 2025: The Fifth Pandust Conference"](#)
 Moderator for evolution of dust properties discussion.
45. Han, J.J. et al. [208 additional co-authors including **Saydjari, A. K.**] 2023, arXiv:2306.11784. Grad
[NANCY: Next-generation All-sky Near-infrared Community survey](#)
 Lead (wrote) the dust/reddening applications section.
46. Paladini, R. et al. [69 additional co-authors including **Saydjari, A. K.**] 2023, arXiv:2307.07642. Grad
[Roman Early-Definition Astrophysics Survey Opportunity: Galactic Roman Infrared Plane Survey \(GRIPS\)](#)
 (Co)-Developed crowded field photometry codes necessary for successful photometry in the plane.
47. Molinari, S. et al. [29 additional co-authors including **Saydjari, A. K.**] 2025, PRIMA GO Science Book Volume 2, 435. Postdoc
[PRIMAGAL, a full Galactic Plane Survey with PRIMAgar: The Role of Magnetic Field in the Formation, Fragmentation and Demise of Dense star-forming Filamentary Clouds](#)
 Improved survey depth/design forecasting
48. Han, J.J. et al. [95 additional co-authors including **Saydjari, A. K.**] 2026, arXiv:2602.21280. Postdoc
[A Path to an All-Sky Survey with Roman](#)
 Based on contributions to NANCY.

Technical Publications as a Contributing Author:

49. Ting, Y.-S. et al. [23 additional co-authors including **Saydjari, A. K.**] 2026, RNAAS, 10, 202. Postdoc
[Deep Learning for Astrophysics: An Open Textbook from the NASA Cosmic Origins AI/ML Science and Technology Interest Group](#)
50. Sayres, C. et al. [12 additional co-authors including **Saydjari, A. K.**] 2022, Proceedings of the SPIE, 12184 Grad
[SDSS-V robotic focal plane system: overview of coordinate systems and transforms](#)

Developed regularized fits to improve focal plane coordinate transform.